SPATIAL TIME SERIES FORECASTING WITH MISSING DATA THROUGH CO-OBSERVATION-GUIDED REGIONAL GRAPH MODELING

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ABSTRACT

Spatial monitoring streams rely on node histories and inter-sensor relations. Asynchronous missing observations across sensors create two coupled challenges: they shorten the available history at individual nodes and reduce the pairwise co-observations used to infer relations for graph message passing. Existing missing-data forecasting methods primarily handle incomplete histories through imputation or node-wise masks, without explicitly assessing whether learned relations are supported by sufficient pairwise evidence. We propose MR-MSGF, a graph forecasting framework that directly predicts future values from incomplete spatiotemporal histories. Before sparse selection, MR-MSGF regularizes window-specific learned relation scores with pairwise co-observation, fuses the resulting graph with physical topology, and aggregates regional context within physically connected regions. Missingness and regional features then gate regional-message injection and bias the multi-scale readout. Compared with the graph-based baseline HD-TTS-AMP, MR-MSGF lowers MAE in all five missingness settings on METR-LA traffic-speed data and also when a high-rate Point mask is additionally applied to AQI36 air-quality data.

***Index Terms*—** Multivariate time series forecasting, missing observations, spatiotemporal graph learning, graph coarsening

1. INTRODUCTION

Traffic, environmental, and industrial forecasting systems depend on temporal context and inter-sensor relations. Spatiotemporal graph models represent these relations with physical topology or learned adjacency [1, 2], while hybrid designs combine physical and data-driven dependencies [3]. Both relation types are used to propagate information across sensors. Under non-synchronous outages, two equally similar node representations can rest on very different amounts of joint observation. Missingness therefore affects not only node histories, but also whether an inferred edge has sufficient evidence for message passing.

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Missing-data research primarily addresses the node-history problem. Imputation reconstructs values through bidirectional recurrent models [4], temporal attention for irregular samples [5], or graph propagation [6, 7]; direct forecasting instead encodes values and masks jointly [8, 9]. These routes make incomplete histories usable for prediction. Once their features enter graph learning, however, node availability alone does not state how much jointly observed history supports a candidate node pair.

Pairwise evidence is distinct from a node-wise missingness indicator. A large adaptive similarity can be estimated from only a few coincident observations, or from a long jointly observed history; its numerical strength is the same type of quantity, but its support is not. The distinction matters before top-k selection commits candidate relations to subsequent graph propagation.

Relation modeling under irregular observation has also been studied beyond forecasting. RAINDROP learns sample-specific sensor graphs from irregular observations for classification [10], while TGNN4I handles partial node observations and irregular times through continuous-time graph states [11]. For missing-data forecasting, BiTGraph injects missing-pattern biases into temporal and graph propagation [12]; GinAR combines predefined and adaptive graphs under variable missingness [13]; and GCN-M studies graph learning with incomplete traffic observations [14]. To our knowledge, these methods model missingness together with node representations or graph construction, but do not explicitly separate a learned relation's strength from the amount of pairwise evidence supporting it.

Multi-scale models further aggregate context across time and space [15, 16, 17]. A physical topology supplies a stable locality prior that does not depend on the current missing window, whereas a learned graph can capture window-specific functional dependence. They are complementary, but neither source alone determines whether a coarse regional message is useful for a particular node. A connected region can still have sparse coverage, long outages, or a node representation far from its regional center. Fusing relations before physically connected regionalization retains both forms of evidence; conditioning the resulting regional context on such node, edge, and regional cues avoids treating coarse aggrega-

tion as universally beneficial.

This leaves a specific question: given incomplete history, how should a model condition learned-edge propagation and induced regional messages on available observation support? The goal is not to recover a ground-truth graph or calibrated uncertainty, but to condition forecasting messages on available evidence. MR-MSGF regularizes learned relations with co-observation, fuses them with physical topology before connected regionalization, and gates regional context with observation and regional features. Our contributions are:

- We use pairwise co-observation as observation support to regularize learned edges before sparse selection.
- We fuse observation-supported and physical graphs, regionalize the fusion, and use observation and regional features to condition regional messages and multi-scale readout.
- METR-LA results are consistent with this bias; AQI36 and extreme outages delimit its scope.

2. METHOD

2.1. Problem Setup and Backbone

Let $X \in \mathbb{R}^{T\times N\times C}$ be a history window, $M \in \{0,1\}^{T\times N\times C}$ its observation mask, and $A^{\mathrm{phy}} \in \mathbb{R}^{N\times N}$ the physical adjacency. The model predicts $Y \in \mathbb{R}^{H\times N\times C_y}$ from $X^{\mathrm{obs}} = M \odot X$, M, and A^{phy}. Graphs and observation features are computed per window; we omit the batch index. All datasets considered here have one target channel per node ($C = C_y = 1$), so $\bar{M}_{t,i} = M_{t,i,1}$ and

$$r_i = \frac{1}{T}\sum_t \bar{M}_{t,i}, \qquad q_{ij} = \frac{1}{T}\sum_t \bar{M}_{t,i}\bar{M}_{t,j}. \tag{1}$$

Here, r_i denotes availability and q_{ij} denotes pairwise co-observation support; all adjacency matrices used for message propagation are row-normalized. We retain the mask-aware encoder, dilated temporal hierarchy, fixed physical spatial hierarchy, and readout of HD-TTS-AMP [15]. This backbone maps incomplete history to level-wise node states $H_\ell \in \mathbb{R}^{N\times d}$ and multi-resolution spatial states $B_{s,\ell}$. Here ℓ indexes temporal levels, $s = 0$ denotes the finest spatial level, and $s > 0$ denotes coarser spatial levels. MR-MSGF replaces only $B_{0,\ell}$ by $\widetilde{B}_{0,\ell}$ through relation fusion and regional propagation; the coarser states remain unchanged and provide complementary context to the multi-scale readout. All $W_q, W_k, W_f, W_s, W_c, W_r, W_p \in \mathbb{R}^{d\times d}$, while $w_{s,\ell} \in \mathbb{R}^4$.

Fig. 1 outlines MR-MSGF: co-observation guides fine relations, whose regional context is controlled by missingness-conditioned features for prediction.

2.2. Observation-Supported Relation Fusion

The encoder produces history features $E \in \mathbb{R}^{T\times N\times d}$. We obtain a window-specific node representation by masked pooling:

$$u_i = \frac{\sum_t \bar{M}_{t,i}E_{t,i}}{\sum_t \bar{M}_{t,i} + \epsilon}. \tag{2}$$

The observation-supported score combines learned similarity and co-observation:

$$s_{ij} = \frac{(W_q u_i)^\top (W_k u_j)}{\sqrt{d}}, \tag{3}$$

$$\tilde{s}_{ij} = s_{ij} + \mathrm{softplus}(\beta)\log(q_{ij} + \epsilon). \tag{4}$$

Here β is learnable. We retain the top-k entries over the full directed score row ($k = 8$), apply row softmax, add one explicit identity contribution, and row-normalize to obtain directed sparse A^{obs}. Thus, a selected diagonal score is a learned candidate contribution, whereas the added identity is the only fixed self-loop contribution. If $r_i = 0$, we set $A^{\mathrm{obs}}_{i,:} = 0$ and replace its fine-graph row with the physical-graph row; this prevents arbitrary top-k ties from creating learned edges for a fully unobserved source. The logarithmic term softly discourages edges with little joint observation before sparsification.

Let $\bar{A}^{\mathrm{phy}}$ denote the physical graph after adding self-loops and applying row normalization. A learned edge gate then forms the fine relation:

$$g_{ij} = \mathrm{sigmoid}\Big(\psi_g([r_i, r_j, q_{ij}, \bar{A}^{\mathrm{phy}}_{ij}, A^{\mathrm{obs}}_{ij}])\Big), \tag{5}$$

$$A^{\mathrm{fine}} = \mathrm{RowNorm}\big(g \odot \bar{A}^{\mathrm{phy}} + (1-g)\odot A^{\mathrm{obs}}\big). \tag{6}$$

Thus, both physical structure and observation-supported relations participate in end-to-end learned relation fusion. The gate is intentionally edge-specific rather than a global mixing coefficient, allowing physical and observation-supported evidence to receive different weights for different edges. In implementation, ψ_g is a two-layer $5 \to 16 \to 1$ MLP with GELU activation.

2.3. Missingness-Conditioned Regional Context

We construct a fixed hard assignment $S \in \{0,1\}^{N\times K}$ from the physical graph, with each node assigned to exactly one region. In METR-LA, A^{phy} is the standard distance connectivity thresholded at 0.1; its directed, row-normalized form is used for propagation. Only for region construction, we symmetrize this adjacency and remove self-loops. We then seed by remaining physical degree and greedily add the unassigned node with the largest positive affinity to a region member until no such node remains or size $m = 8$ is reached. Thus regions are physically connected and do not absorb zero-affinity

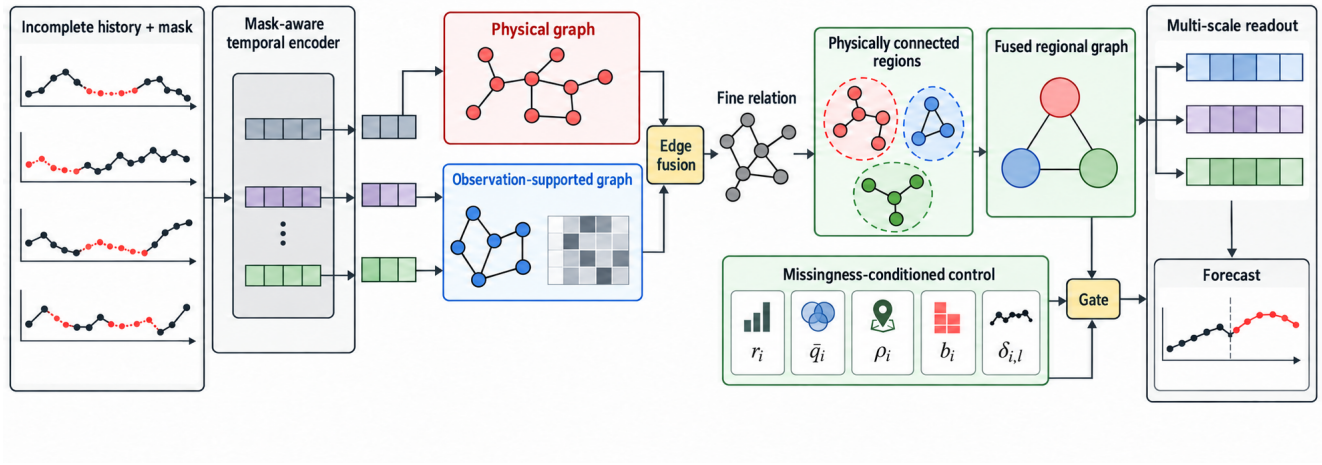


Fig. 1. Overview of MR-MSGF. A mask-aware encoder forms temporal states from incomplete history. A co-observation-supported graph is fused with the physical graph before physically connected regional coarsening. Observation and regional features regulate regional residual injection and multi-scale readout.

nodes. With $n_k = \sum_i S_{ik}$,

$$Z_\ell^{\text{fine}} = \text{ELU}(A^{\text{fine}} H_\ell W_f + H_\ell W_s), \tag{7}$$

$$H_{\ell,k}^{\text{reg}} = \frac{1}{n_k} \sum_i S_{ik} Z_{\ell,i}^{\text{fine}}, \tag{8}$$

$$A^{\text{coarse}} = \text{RowNorm}(S^\top A^{\text{fine}} S). \tag{9}$$

Regional propagation and lifting are

$$Z_\ell^{\text{reg}} = \text{ELU}(A^{\text{coarse}} H_\ell^{\text{reg}} W_c + H_\ell^{\text{reg}} W_r), \tag{10}$$

$$Z_\ell^{\text{coarse}} = S Z_\ell^{\text{reg}}. \tag{11}$$

Coarsening after fine-scale fusion lets regional aggregation operate on a graph containing both physical structure and observation-supported relations; coarsening only the physical or observation graph cannot retain both sources of evidence. $S^\top A^{\text{fine}} S$ sums fine-edge weights between regions, so A^{coarse} represents total inter-region edge mass rather than average pairwise affinity; the final row normalization converts these totals into outgoing regional propagation weights.

For node i, let $\bar{q}_i = \sum_j \bar{A}_{ij}^{\text{phy}} q_{ij}$ be physical-neighbor-weighted co-observation, $\rho_i = \sum_k S_{ik} n_k^{-1} \sum_j S_{jk} r_j$ be regional coverage, and b_i be the longest missing block in the current history window divided by T. Let $k(i)$ denote the region of node i, $c_{i,\ell} = H_{\ell,k(i)}^{\text{reg}}$, and

$$\zeta_{i,\ell} = \sqrt{\max\left(d^{-1} \| Z_{\ell,i}^{\text{fine}} - c_{i,\ell} \|_2^2, \epsilon\right)}, \tag{12}$$

$$\delta_{i,\ell} = \frac{\zeta_{i,\ell}}{N^{-1} \sum_j \zeta_{j,\ell} + \epsilon}. \tag{13}$$

Here $\delta_{i,\ell}$ is normalized representation dispersion from the regional center. The missingness-conditioned regional gate is

$$\alpha_{i,\ell} = \text{sigmoid}\left(\psi_\alpha([r_i, \bar{q}_i, \rho_i, b_i, \delta_{i,\ell}]) \right. \tag{14}$$

$$\left. + \gamma^\top [1 - r_i, 1 - \bar{q}_i, \rho_i - 1, b_i, -\delta_{i,\ell}]\right). \tag{15}$$

Here ψ_α uses the same two-layer MLP structure as the edge gate, and γ is a learnable scaling vector. The second term is a residual feature vector containing signed complementary observation and regional-consistency cues; since γ is not sign-constrained, we do not interpret it as a monotonic prior. Let $\boldsymbol{\alpha}_\ell = [\alpha_{1,\ell}, \ldots, \alpha_{N,\ell}]^\top \in \mathbb{R}^N$, broadcast along the feature dimension. The finest representation becomes

$$\widetilde{B}_{0,\ell} = Z_\ell^{\text{fine}} + \boldsymbol{\alpha}_\ell \odot \left(Z_\ell^{\text{coarse}} W_p\right), \tag{16}$$

while $\widetilde{B}_{s,\ell} = B_{s,\ell}$ for $s > 0$. We further bias multi-scale readout by observation and regional features. Let $R_i = [r_i, \bar{q}_i, \rho_i, b_i]^\top$. The score for prediction step h is

$$e_{h,s,\ell,i} = e_{h,s,\ell,i}^{\text{repr}} + w_{s,\ell}^\top R_i. \tag{17}$$

The parameters $w_{s,\ell}$ are shared across prediction steps; horizon dependence enters through $e_{h,s,\ell,i}^{\text{repr}}$. For each node and prediction step, a softmax over (s, ℓ) yields $\pi_{h,s,\ell,i}$, and $F_{h,i} = \sum_{s,\ell} \pi_{h,s,\ell,i} \widetilde{B}_{s,\ell,i}$ is mapped to $\hat{Y}_{h,i}$ by the prediction head. Thus, these features bias scale selection rather than replacing representation content. Training minimizes masked forecasting MAE,

$$\mathcal{L}_{\text{pred}} = \frac{\sum M^Y \odot |\hat{Y} - Y|}{\sum M^Y}, \tag{18}$$

where M^Y excludes unavailable future targets. We do not use reconstruction, graph, or clustering auxiliary losses. The gate is not interpreted as calibrated uncertainty. It is a forecasting-oriented control variable that decides whether regional context should be added for a node at a temporal level. This design keeps the added branch lightweight while making regional messages conditional on missingness evidence rather than unconditional smoothing.

3. EXPERIMENTS

3.1. Settings

We evaluate METR-LA traffic speeds and AQI36 air quality, forecasting 12 steps from 24 histories. METR-LA Point and Block-T masks affect only history; valid, unmasked raw targets are evaluated. Point simulates independent loss and Block-T contiguous outages. AQI36 retains its native validity mask with optional Point masks. Conditions use five paired mask seeds and shared input masks. Training uses only Eq. (18). We use $k = m = 8$, initialize $\beta = 1$ and γ to ones, and obtain $K = 31$ and $K = 5$ connected regions for METR-LA and AQI36. HD-TTS-AMP is the main reference; BiTGraph uses the common data, mask, and evaluation interface.

3.2. Main Results

Table 1 reports METR-LA results. MR-MSGF lowers MAE over HD-TTS-AMP in all tested Block-T and Point settings. It also improves RMSE under Block-T 30%/50% and Point 30%/50%. At Block-T 70%, MR-MSGF still improves MAE but has slightly worse RMSE, indicating that the benefit is not cross-metric consistent under extreme missingness. Across all five conditions, MR-MSGF wins all five paired MAE runs and two-sided paired t-tests give $p \leq 0.0379$; we make no analogous across-condition significance claim for RMSE.

Table 1. METR-LA results over five paired mask seeds. Lower is better; the best result within each condition is in bold.

Mask	Rate	Method	MAE	RMSE
Block-T	30%	HD-TTS-AMP	$3.228 \pm .011$	$6.604 \pm .013$
		BiTGraph	$3.823 \pm .011$	$7.884 \pm .037$
		MR-MSGF	$\mathbf{3.200 \pm .016}$	$\mathbf{6.584 \pm .049}$
	50%	HD-TTS-AMP	$3.355 \pm .013$	$6.878 \pm .022$
		BiTGraph	$3.973 \pm .018$	$8.191 \pm .046$
		MR-MSGF	$\mathbf{3.308 \pm .020}$	$\mathbf{6.815 \pm .055}$
	70%	HD-TTS-AMP	$3.465 \pm .010$	$\mathbf{7.093 \pm .042}$
		BiTGraph	$4.136 \pm .013$	$8.481 \pm .031$
		MR-MSGF	$\mathbf{3.446 \pm .010}$	$7.109 \pm .038$
Point	30%	HD-TTS-AMP	$3.105 \pm .007$	$6.359 \pm .015$
		BiTGraph	$3.637 \pm .012$	$7.526 \pm .038$
		MR-MSGF	$\mathbf{3.076 \pm .012}$	$\mathbf{6.309 \pm .045}$
	50%	HD-TTS-AMP	$3.160 \pm .011$	$6.476 \pm .022$
		BiTGraph	$3.694 \pm .005$	$7.658 \pm .029$
		MR-MSGF	$\mathbf{3.141 \pm .007}$	$\mathbf{6.456 \pm .029}$

Table 2. AQI36 cross-domain results over five seeds.

Condition	Method	MAE	RMSE
Native	HD-TTS-AMP	$\mathbf{30.484 \pm .585}$	$\mathbf{49.333 \pm 1.407}$
	MR-MSGF	$30.727 \pm .239$	$49.888 \pm .929$
Native+P25	HD-TTS-AMP	$31.203 \pm .689$	50.371 ± 1.035
	MR-MSGF	$\mathbf{31.117 \pm .416}$	$\mathbf{50.333 \pm .782}$
Native+P50	HD-TTS-AMP	$31.736 \pm .496$	$\mathbf{50.403 \pm 1.204}$
	MR-MSGF	$\mathbf{31.645 \pm .709}$	50.527 ± 1.386

3.3. Cross-Domain Analysis

AQI36 provides a cross-domain stress test with native missingness. Table 2 shows that MR-MSGF is worse than HD-TTS-AMP under native missingness, improves both metrics with an additional 25% Point mask, and lowers MAE but raises RMSE with an additional 50% Point mask. The heterogeneous native missingness need not make the current observation features align with transferable relations; this is a hypothesis suggested by the result, not a causal explanation. We therefore treat AQI36 as a boundary condition rather than claiming dataset-independent gains.

3.4. Ablation and Diagnostics

Table 3 reports Block-T 50% component sensitivity. The full model is best on both metrics. Removing the co-observation score penalty, the missingness-conditioned regional gate, or the readout bias increases mean error, and using only one fine relation source is weaker than the fused graph. These mean patterns are consistent with complementary roles for co-observation-guided fusion and conditional regional context, but do not isolate causal effects of individual components.

4. CONCLUSION

MR-MSGF forecasts incomplete spatial histories by regularizing learned graph propagation with pairwise co-observation, fusing it with physical structure, and conditioning regional context on missingness and regional features. Paired METR-LA results are consistent with this observation-aware inductive bias, while Block-T 70% and AQI36 delimit its cross-metric and cross-domain scope.

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Table 3. METR-LA Block-T 50% ablation over five paired seeds.

Variant	MAE	RMSE
HD-TTS-AMP	3.355 ± 0.013	6.878 ± 0.022
Physical only	3.329 ± 0.012	6.861 ± 0.049
Observation only	3.335 ± 0.016	6.887 ± 0.059
Fused graph	3.318 ± 0.011	6.829 ± 0.046
No q_{ij} penalty	3.315 ± 0.020	6.833 ± 0.055
No regional gate	3.314 ± 0.011	6.840 ± 0.042
No readout bias	3.330 ± 0.025	6.851 ± 0.038
Full **MR-MSGF**	$\mathbf{3.308 \pm 0.020}$	$\mathbf{6.815 \pm 0.055}$

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